

# Roof Lifecycle Cost Analysis

Planned maintenance is projected to save the owner of a typical Hampton Roads mixed-use building **\$277K** over the next 20 years.

About 32% less than reactive repair: \$194K lower spending plus \$83K more roof life at year 20. Range \$135K–\$367K from conservative to upside assumptions; \$145K in today's dollars at a 7% discount rate.

| MARKET        | PROPERTY  | ROOF AREA    | SYSTEM | AGE · CONDITION | NEW ROOF TODAY |
|---------------|-----------|--------------|--------|-----------------|----------------|
| Hampton Roads | Mixed Use | 25,000 sq ft | TPO    | 8 yrs · Good    | \$225K         |

**PAXSEAL NET COST**

**\$588K**

\$565K spent + \$135K roof used – \$113K roof left

**REACTIVE NET COST**

**\$865K**

\$760K spent + \$135K roof used – \$30K roof left

**VALUE PROTECTED**

**\$159K**

\$11K/yr more NOI at a 6.6% cap rate, if the owner pays for the roof

**ROOF PRICES**

**2× in 15 yrs**

At 4.6%/yr, the long-run trend in commercial roofing costs

## Cumulative roof spending, next 20 years

Nominal dollars. Vertical jumps are full roof replacements.



## Roof life

When each scenario needs a new roof.



Where the savings come from

20-year totals

|                           | PAXSEAL | REACTIVE | SAVINGS |
|---------------------------|---------|----------|---------|
| PaxSeal fees              | \$109K  | —        | -\$109K |
| Repairs paid by owner     | \$55K   | \$388K   | +\$333K |
| Expected storm damage     | \$33K   | \$49K    | +\$16K  |
| Roof replacements         | \$369K  | \$322K   | -\$47K  |
| Cash spending             | \$565K  | \$760K   | +\$195K |
| Current roof value used   | \$135K  | \$135K   | —       |
| Roof life left at year 20 | -\$112K | -\$30K   | +\$82K  |
| Net 20-year cost          | \$588K  | \$865K   | +\$277K |

What this means

- 1 **A later replacement.** The next new roof moves from year 8 to year 11, leaving 10 years of roof life at year 20 instead of 2.
- 2 **Smaller repairs.** Emergency repairs run about 3.6× the planned rate in a mixed-use building, and the PaxSeal fee covers half of planned repairs. Together that saves \$333K.
- 3 **Both income streams protected.** A leak can close the ground-floor retail and damage the offices or apartments above.
- 4 **The program pays for itself.** Each \$1 of fees avoids about \$2.78 of other cash roof costs.

Local risk profile

Norfolk city, VA (51710)

Hurricane / tropical-storm wind

About 4.8 tropical storms or stronger within 50 nautical miles of Norfolk per 20 years (NOAA HURDAT2 1975–2024). Expected storm damage to this roof over 20 years is **\$33K** if maintained and **\$49K** with reactive repair only, before insurance. A documented maintenance history also helps when an insurer reviews a claim.

|                                 |                     |       |
|---------------------------------|---------------------|-------|
| Hail                            | Relatively Low      | ■■■■■ |
| Strong wind                     | Relatively Moderate | ■■■■■ |
| Tornado                         | Relatively Moderate | ■■■■■ |
| Hurricane                       | Relatively Moderate | ■■■■■ |
| Winter weather                  | Relatively Moderate | ■■■■■ |
| Ice storm                       | Relatively Moderate | ■■■■■ |
| FEMA National Risk Index rating |                     |       |

Key assumptions

The basis for each figure in this analysis.

| ASSUMPTION           | VALUE            | BASIS                                                                                                     |
|----------------------|------------------|-----------------------------------------------------------------------------------------------------------|
| Roof life            | 20 / 15 yrs      | Maintained / reactive. Industry life tables; maintenance adds ~5 yrs (NRC Canada, warranties)             |
| Roof cost growth     | 4.6%/yr          | BLS Producer Price Index, nonresidential roofing, 2007–2025                                               |
| Replacement cost     | \$9.00/sq ft     | Hampton Roads contractor price ranges                                                                     |
| Reactive repair cost | 3× planned × 1.2 | Industry estimates; ×1.2 for leak damage (mixed-use building). PaxSeal fee covers half of planned repairs |
| Storm damage         | 0.44%/yr         | Of roof cost, before insurance; ×1.5 if reactive. FEMA National Risk Index                                |
| PaxSeal fee          | \$4,050/yr       | Year 1, rising 3%/yr. Estimate; your proposal states the actual fee                                       |
| Cap rate             | 6.6%             | PAX regional median                                                                                       |

**Next step:** a roof assessment and a written PaxSeal proposal for this building. Call PAX Services Group at 1-855-729-7510 or email [info@paxserv.com](mailto:info@paxserv.com).

**Estimates, not a quote.** Projections from Hampton Roads market research and industry data, in nominal dollars. The PaxSeal fee is an estimate; your proposal states the actual price. Net cost includes wear on the current roof, less roof life left at year 20; both roof-value figures are in today's dollars. Storm damage is before insurance. Property value is the yearly gain in net operating income divided by the cap rate, for income property where the owner pays for the roof.